11-matching-flow: 匹配与网络流 (Matching and Network Flow)

魏恒峰

hfwei@hnu.edu.cn

2026 年 05 月 26 日

3 Theorems + 1 Algorithm

To maximize the size of a mathematical structure $\mathcal{S}$ in G

To maximize the size of a mathematical structure $\mathcal{S}$ in G

Theorem

$\mathcal{S}$ is maximum iff G does not contain $\mathcal{S}$-augmenting objects.

To maximize the size of a mathematical structure $\mathcal{S}$ in G

Theorem

$\mathcal{S}$ is maximum iff G does not contain $\mathcal{S}$-augmenting objects.

Algorithm

Repeatedly finding $\mathcal{S}$-augmenting objects until no more ones exist.

To maximize the size of a mathematical structure $\mathcal{S}$ in G

To maximize the size of a mathematical structure $\mathcal{S}$ in G

To minimize the size of its dual mathematical structure $\mathcal{S}'$ in G

To maximize the size of a mathematical structure $\mathcal{S}$ in G

To minimize the size of its dual mathematical structure $\mathcal{S}'$ in G

Theorem (Weak Duality Theorem)

The size of a maximum $\mathcal{S}$ $\leq$ *The size of a minimum* $\mathcal{S}'$

To maximize the size of a mathematical structure $\mathcal{S}$ in G

To minimize the size of its dual mathematical structure $\mathcal{S}'$ in G

Theorem (Weak Duality Theorem)

The size of a maximum $\mathcal{S} \leq$ *The size of a minimum* $\mathcal{S}'$

Theorem (Strong Duality Theorem)

The size of a maximum $\mathcal{S} =$ *The size of a minimum* $\mathcal{S}'$

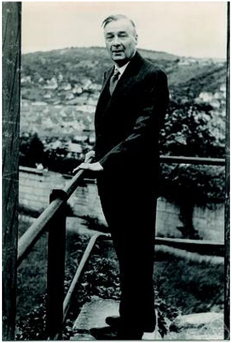

let's get
married
today

The Marriage Problem (Philip Hall, 1935)

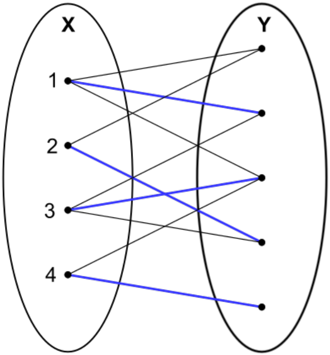
If there is a finite set of girls, each of whom knows several boys, under what conditions can all the girls marry boys in such a way that each girl marries a distinct boy that she knows?

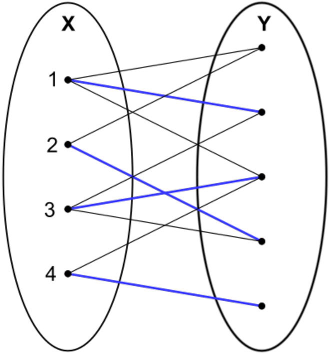
The Marriage Problem (Philip Hall, 1935)

If there is a finite set of girls, each of whom knows several boys, under what conditions can all the girls marry boys in such a way that each girl marries a distinct boy that she knows?



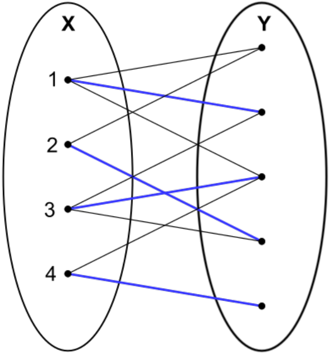
Philip Hall (1904 ∼ 1982)

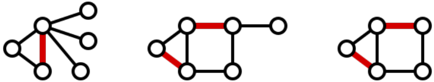

X
Y
1
2
3
4



Definition (Matching (匹配))

A matching in a graph G is a set of edges with no shared endpoints.

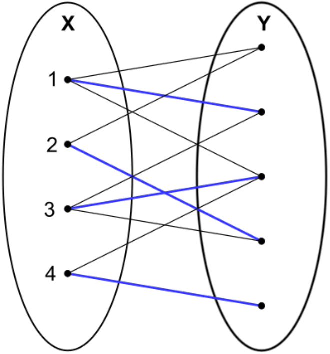


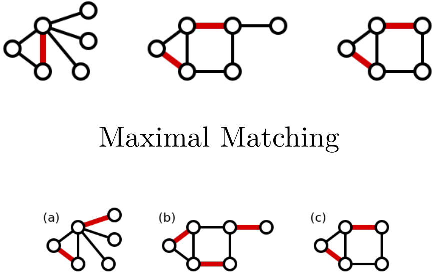


Maximal Matching

Definition (Matching (匹配))

A matching in a graph G is a set of edges with no shared endpoints.



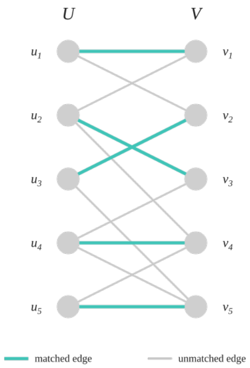


Maximal Matching

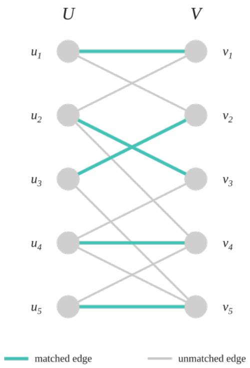
Maximum Matching

Definition (Matching (匹配))

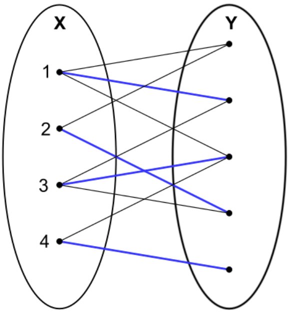
A matching in a graph G is a set of edges with no shared endpoints.



Perfect Matching



Perfect Matching



X-Perfect Matching

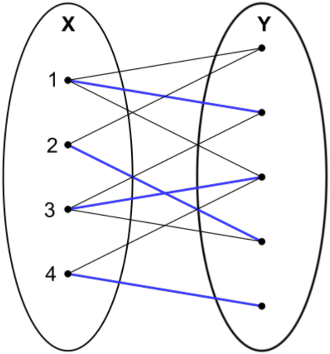
Definition (X-Perfect Matching (X-Saturating Matching))

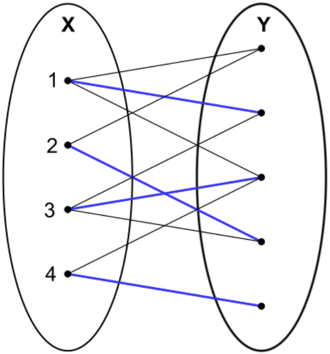
Let $G = (X, Y, E)$ be a bipartite graph.

An X-perfect matching of G is a matching which covers each vertex in X.

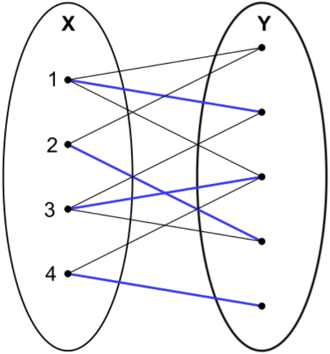
The Marriage Problem (Philip Hall, 1935)

Let $G = (X, Y, E)$ be a bipartite graph.
Is there an X-perfect matching of G?

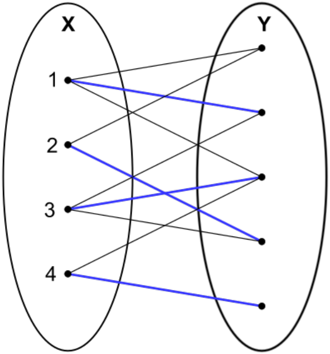



X
Y
1
2
3
4

$$|X| \leq |Y|$$



$$\left|X\right| \le \left|Y\right|$$



$$\forall W \subseteq X.\ \left|W\right| \le \left|N(W)\right|$$

Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph.

There is an X-perfect matching of G if and only if

$$\forall W \subseteq X \; \left|W\right| \le \left|N_G(W)\right|.$$

By induction on the number $\left|X\right|$ of vertices in X.

By induction on the number $\left|X\right|$ of vertices in X.

Basis Step: $\left|X\right| = 1$. $\left|X\right| \leq \left|N_G(X)\right|$. I am married!

By induction on the number $\left|X\right|$ of vertices in X.

Basis Step: $\left|X\right| = 1$. $\left|X\right| \leq \left|N_G(X)\right|$. I am married!

Induction Hypothesis: Suppose that it holds if $\left|X\right| < m$.

By induction on the number $\left|X\right|$ of vertices in X.

Basis Step: $\left|X\right| = 1$. $\left|X\right| \leq \left|N_G(X)\right|$. I am married!

Induction Hypothesis: Suppose that it holds if $\left|X\right| < m$.

Induction Step: Consider the case $\left|X\right| = m$.

By induction on the number $\left|X\right|$ of vertices in X.

Basis Step: $\left|X\right| = 1$. $\left|X\right| \leq \left|N_G(X)\right|$. I am married!

Induction Hypothesis: Suppose that it holds if $\left|X\right| < m$.

Induction Step: Consider the case $\left|X\right| = m$.

By induction on the number $\left|X\right|$ of vertices in X.

Basis Step: $\left|X\right| = 1$. $\left|X\right| \le \left|N_G(X)\right|$. I am married!

Induction Hypothesis: Suppose that it holds if $\left|X\right| < m$.

Induction Step: Consider the case $\left|X\right| = m$.

How to reduce the case $\left|X\right| = m$ to a case $\left|X'\right| < m$?

Consider the case $\left|X\right| = m$.

Consider the case $\left|X\right| = m$.

- CASE I: Every $k < m$ girls in X know $\geq k+1$ boys in Y.

Consider the case $\left|X\right| = m$.

- CASE I: Every $k < m$ girls in X know $\geq k+1$ boys in Y.

  Take any girl x and marry her to any boy y she knows.

$$\text{Consider the case } \left|X\right| = m.$$

- CASE I: Every $k < m$ girls in X know $\geq k+1$ boys in Y.

  Take any girl x and marry her to any boy y she knows.

$$G' = G - \{x, y\}$$

Consider the case $\left|X\right| = m$.

- CASE I: Every $k < m$ girls in X know $\geq k+1$ boys in Y.

  Take any girl x and marry her to any boy y she knows.

$$G' = G - \{x, y\}$$

Hall's Condition still holds for G'.

Consider the case $\left|X\right| = m$.

- CASE I: Every $k < m$ girls in X know $\geq k+1$ boys in Y.

Take any girl x and marry her to any boy y she knows.

$$G' = G - \{x, y\}$$

Hall's Condition still holds for G'.

$$\forall W \subseteq X - \{x\}. \ \left|W\right| \leq \left|N_{G'}(W)\right|$$

Consider the case $\left|X\right| = m$.

- CASE I: Every $k < m$ girls in X know $\geq k+1$ boys in Y.

Take any girl x and marry her to any boy y she knows.

$$G' = G - \{x, y\}$$

Hall's Condition still holds for G'.

$$\forall W \subseteq X - \{x\}.\ \left|W\right| \leq \left|N_{G'}(W)\right|$$

There is an $(X - \{x\})$-perfect matching in G'.

Consider the case $\left|X\right| = m$.

- Case I: Every $k < m$ girls in X know $\geq k+1$ boys in Y.

Take any girl x and marry her to any boy y she knows.

$$G' = G - \{x, y\}$$

Hall's Condition still holds for G'.

$$\forall W \subseteq X - \{x\}.\ \left|W\right| \leq \left|N_{G'}(W)\right|$$

There is an $(X - \{x\})$-perfect matching in G'.

Therefore, there is an X-perfect matching in G.

- CASE II: There is a set of $k < m$ girls in X who know exactly k boys in Y.

- Case II: There is a set of $k < m$ girls in X who know exactly k boys in Y.

There k girls can be married by induction to the k boys.

- CASE II: There is a set of $k < m$ girls in X who know exactly k boys in Y.

There k girls can be married by induction to the k boys.

Hall's Condition holds for these k girls and the k boys.

- Case II: There is a set of $k < m$ girls in X who know exactly k boys in Y.

There k girls can be married by induction to the k boys.

Hall's Condition holds for these k girls and the k boys.

$$G' = G - \{\text{these } k \text{ girls}\} - \{\text{these } k \text{ boys}\}$$

- CASE II: There is a set of $k < m$ girls in X who know exactly k boys in Y.

There k girls can be married by induction to the k boys.

Hall's Condition holds for these k girls and the k boys.

$$G' = G - \{\text{these } k \text{ girls}\} - \{\text{these } k \text{ boys}\}$$

Hall's Condition still holds for G'.

- CASE II: There is a set of $k < m$ girls in X who know exactly k boys in Y.

There k girls can be married by induction to the k boys.

Hall's Condition holds for these k girls and the k boys.

$$G' = G - \{\text{these } k \text{ girls}\} - \{\text{these } k \text{ boys}\}$$

Hall's Condition still holds for G'.

By contradition:

Suppose that in G', there are $l \leq m - k$ girls who know $< l$ boys.

- Case II: There is a set of $k < m$ girls in X who know exactly k boys in Y.

There k girls can be married by induction to the k boys.

Hall's Condition holds for these k girls and the k boys.

$$G' = G - \{\text{these } k \text{ girls}\} - \{\text{these } k \text{ boys}\}$$

Hall's Condition still holds for G'.

By contradition:

Suppose that in G', there are $l \le m - k$ girls who know $< l$ boys.

Then, in G, there are $k + l$ girls who know $< k + l$ boys.

- CASE II: There is a set of $k < m$ girls in X who know exactly k boys in Y.

There k girls can be married by induction to the k boys.

Hall's Condition holds for these k girls and the k boys.

$$G' = G - \{\text{these } k \text{ girls}\} - \{\text{these } k \text{ boys}\}$$

Hall's Condition still holds for G'.

By contradition:

Suppose that in G', there are $l \le m - k$ girls who know $< l$ boys.

Then, in G, there are $k + l$ girls who know $< k + l$ boys.

There is an $(X - \{\text{these } k \text{ girls}\})$-perfect matching in G'.

- CASE II: There is a set of $k < m$ girls in X who know exactly k boys in Y.

There k girls can be married by induction to the k boys.

Hall's Condition holds for these k girls and the k boys.

$$G' = G - \{\text{these } k \text{ girls}\} - \{\text{these } k \text{ boys}\}$$

Hall's Condition still holds for G'.

By contradition:

Suppose that in G', there are $l \leq m - k$ girls who know $< l$ boys.

Then, in G, there are $k + l$ girls who know $< k + l$ boys.

There is an $(X - \{\text{these } k \text{ girls}\})$-perfect matching in G'.

Therefore, there is an X-perfect matching in G.

Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph.

$$\forall W \subseteq X \ \left|W\right| \leq \left|N_G(W)\right|.$$

Definition (M-alternating Paths)

Let M be a matching. An M-alternating path is a path that alternates between edges in M and edges not in M.

Definition (M-alternating Paths)

Let M be a matching. An M-alternating path is a path that alternates between edges in M and edges not in M.

Definition (M-augmenting Paths)

An M-augmenting path is an M-alternating path whose endpoints are unsaturated by M.

Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph.

There is a X-perfect matching of G if and only if

$$\forall W \subseteq X \; \left|W\right| \leq \left|N_G(W)\right|.$$

By contradiction.

By contradiction.

Suppose that there is *no* X-perfect matching in G.

By contradiction.

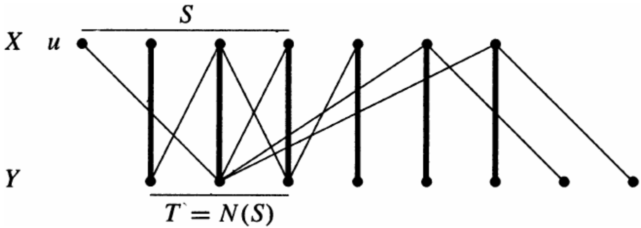
Suppose that there is *no* X-perfect matching in G.

Let M be a *maximum* matching in G.

By contradiction.

Suppose that there is *no* X-perfect matching in G.

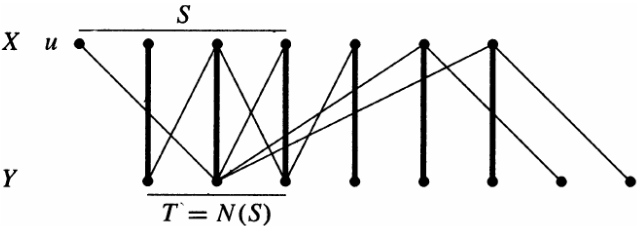
Let M be a *maximum* matching in G.



By contradiction.

Suppose that there is *no* X-perfect matching in G.

Let M be a *maximum* matching in G.

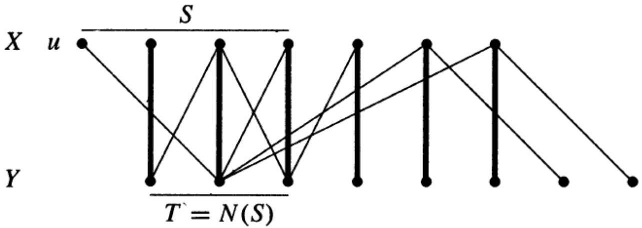


We show that Hall's Condition is violated for some $S \subseteq X$.

By contradiction.

Suppose that there is *no* X-perfect matching in G.

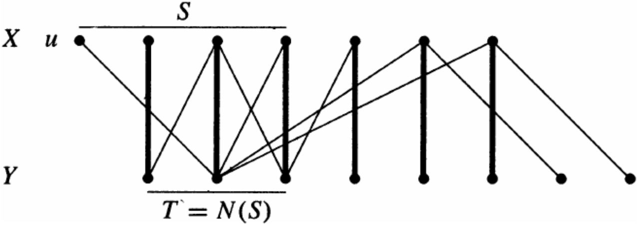
Let M be a *maximum* matching in G.



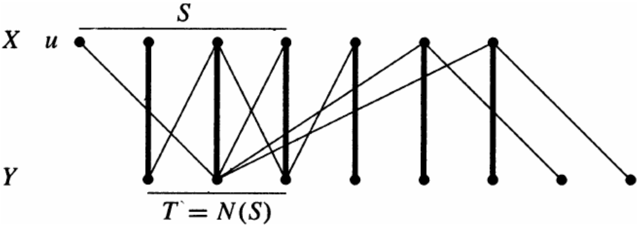
We show that Hall's Condition is violated for some $S \subseteq X$.

We will construct $S \subseteq X$ based on M.

Let $u \in X$ be a vertex of X *not* saturated by M.

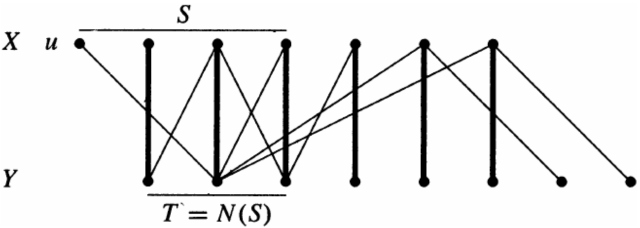


Let $u \in X$ be a vertex of X *not* saturated by M.



Consider all *M-alternating paths* starting from u.

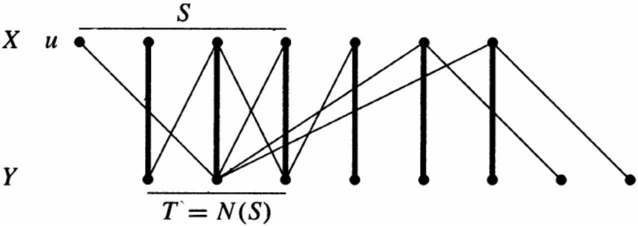
Let $u \in X$ be a vertex of X *not* saturated by M.



Consider all *M-alternating paths* starting from u.

$T \triangleq$ the set of vertices in Y reachable from u by M-alternating paths.

Let $u \in X$ be a vertex of X *not* saturated by M.

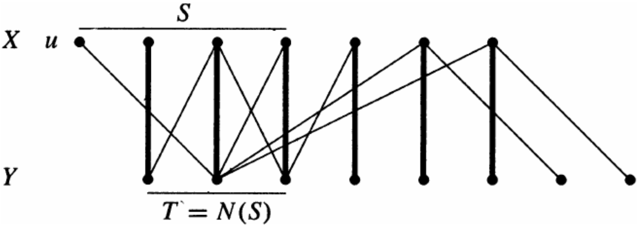


Consider all *M-alternating paths* starting from u.

$T \triangleq$ the set of vertices in Y reachable from u by M-alternating paths.

$S \triangleq$ the set of vertices in X reachable from u by M-alternating paths.

Let $u \in X$ be a vertex of X *not* saturated by M.



Consider all *M-alternating paths* starting from u.

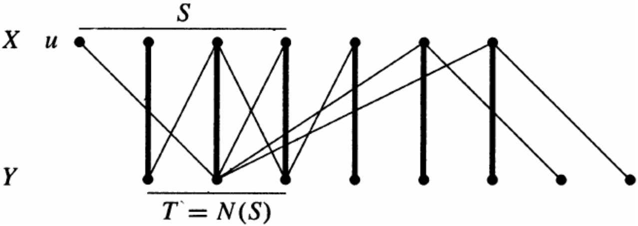
$T \triangleq$ the set of vertices in Y reachable from u by M-alternating paths.

$S \triangleq$ the set of vertices in X reachable from u by M-alternating paths.

We will show that

$$T = N(S) \wedge \Big|T\Big| = \Big|S - \{u\}\Big|$$

Let $u \in X$ be a vertex of X *not* saturated by M.



Consider all *M-alternating paths* starting from u.

$T \triangleq$ the set of vertices in Y reachable from u by M-alternating paths.

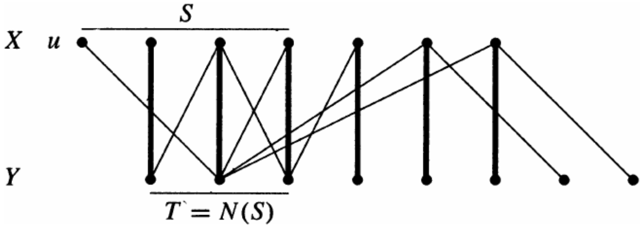
$S \triangleq$ the set of vertices in X reachable from u by M-alternating paths.

We will show that

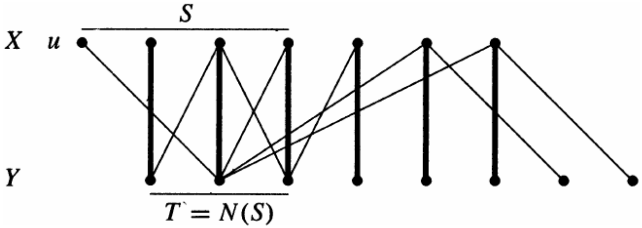
$$T = N(S) \wedge \left|T\right| = \left|S - \{u\}\right|$$

$$\left|N(S)\right| = \left|T\right| = \left|S\right| - 1 < \left|S\right|$$

$$|T| = |S - \{u\}|$$

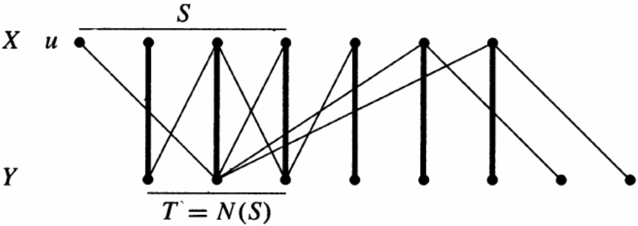


$$\left|T\right| = \left|S - \{u\}\right|$$



We show that there is a bijection from T to $S - \{u\}$.

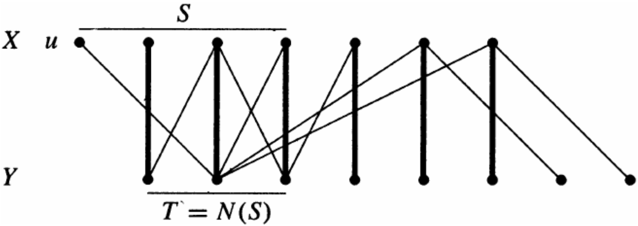
$$|T| = |S - \{u\}|$$



We show that there is a bijection from T to $S - \{u\}$.

M matches T with $S - \{u\}$.

$$|T| = |S - \{u\}|$$

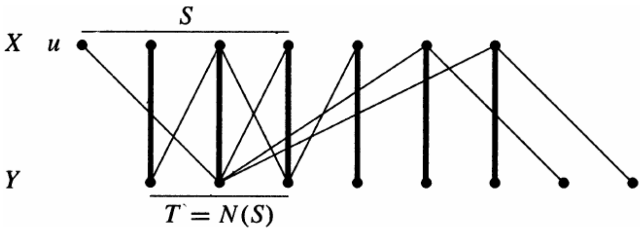


We show that there is a bijection from T to $S - \{u\}$.

M matches T with $S - \{u\}$.

Every vertex in T goes via an edge in M to a vertex in $S - \{u\}$.

$$\left|T\right| = \left|S - \{u\}\right|$$



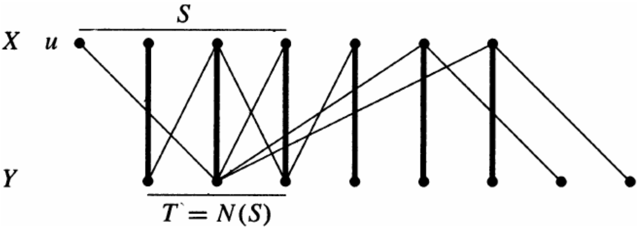
We show that there is a bijection from T to $S - \{u\}$.

M matches T with $S - \{u\}$.

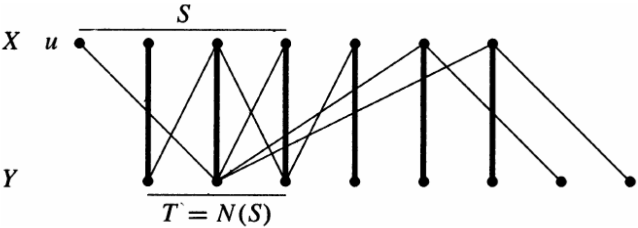
Every vertex in T goes via an edge in M to a vertex in $S - \{u\}$.

Every vertex in $S - \{u\}$ is reached by an edge in M from a vertex in T.

$$T = N(S)$$

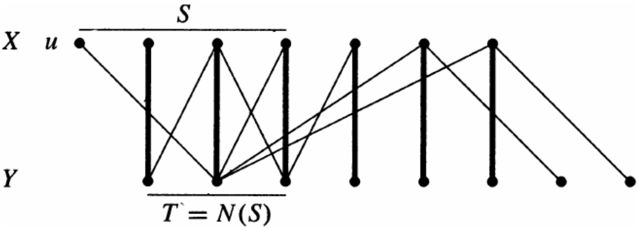


$$T = N(S)$$



$$M \text{ matches } T \text{ with } S - \{u\} \implies T \subseteq N(S)$$

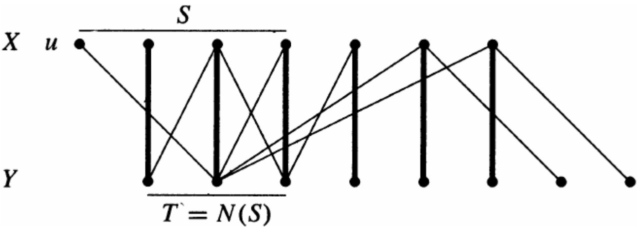
$$T = N(S)$$



$$M \text{ matches } T \text{ with } S - \{u\} \implies T \subseteq N(S)$$

We need to show that $N(S) \subseteq T$

$$T = N(S)$$

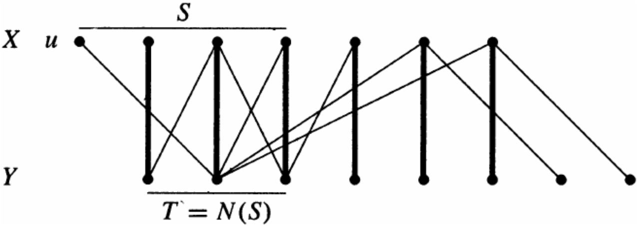


M matches T with $S - \{u\} \implies T \subseteq N(S)$

We need to show that $N(S) \subseteq T$

Suppose by contradiction that $v \in S$ has a neighbor $y \in Y - T$.

$$T = N(S)$$



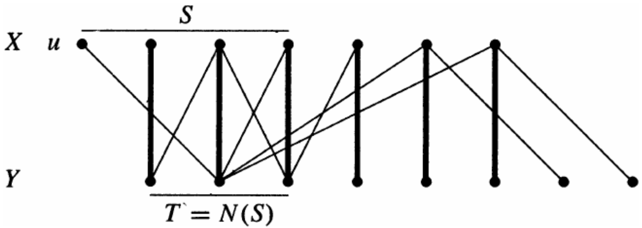
$$M \text{ matches } T \text{ with } S - \{u\} \implies T \subseteq N(S)$$

We need to show that $N(S) \subseteq T$

Suppose by contradiction that $v \in S$ has a neighbor $y \in Y - T$.

It must be $vy \notin M$.

$$T = N(S)$$



M matches T with $S - \{u\} \implies T \subseteq N(S)$

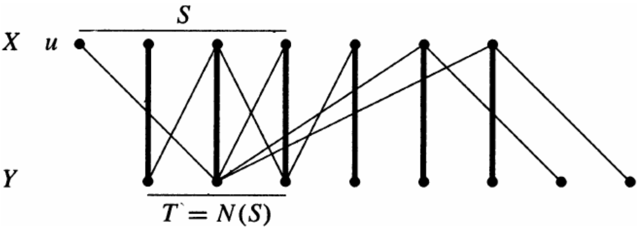
We need to show that $N(S) \subseteq T$

Suppose by contradiction that $v \in S$ has a neighbor $y \in Y - T$.

It must be $vy \notin M$.

There exists an M-alternating path to y from u.

$$T = N(S)$$



M matches T with $S - \{u\} \implies T \subseteq N(S)$

We need to show that $N(S) \subseteq T$

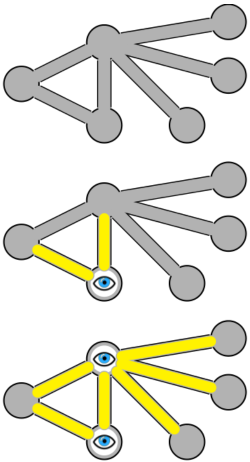
Suppose by contradiction that $v \in S$ has a neighbor $y \in Y - T$.

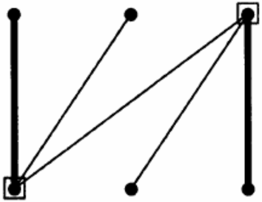
It must be $vy \notin M$.

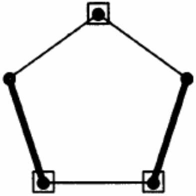
There exists an M-alternating path to y from u. $\implies y \in T$.

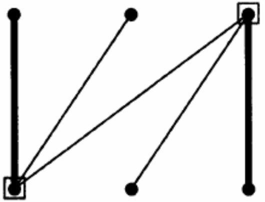
Definitions (Vertex Cover (点覆盖))

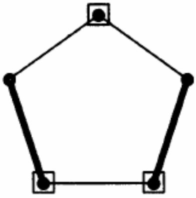
A vertex cover of a graph G is a set $Q \subseteq V(G)$ that covers all edges.

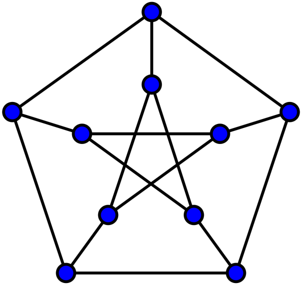










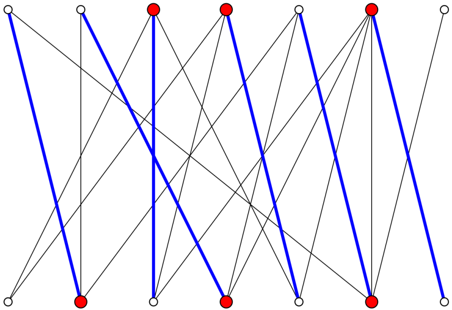


Theorem (Weak Duality Theorem (弱对偶定理))

Let G be a graph. The maximum size of a mathching in G $\leq$ the minimum size of a vertex cover of G.

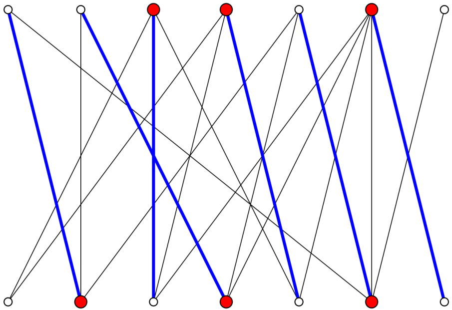
Theorem (Weak Duality Theorem (弱对偶定理))

Let G be a graph. The maximum size of a mathching in G
$\leq$ the minimum size of a vertex cover of G.



Theorem (Weak Duality Theorem (弱对偶定理))

Let G be a graph. The maximum size of a mathching in G $\leq$ the minimum size of a vertex cover of G.

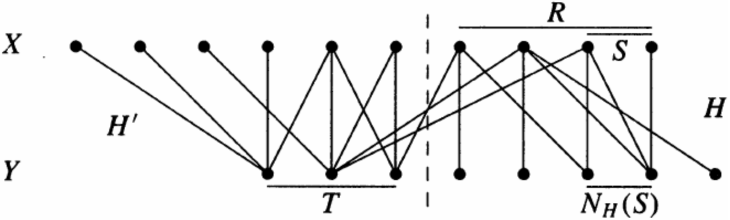


Theorem (König (1931), Egerváry (1931))

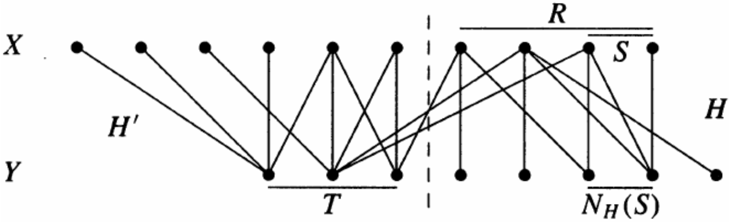
Let G be a ***bipartite*** *graph. The maximum size of a mathching in G $=$ the minimum size of a vertex cover of G*

Given a minimum vertex cover Q,
we construct a matching M of the same size.

Given a minimum vertex cover Q,
we construct a matching M of the same size.

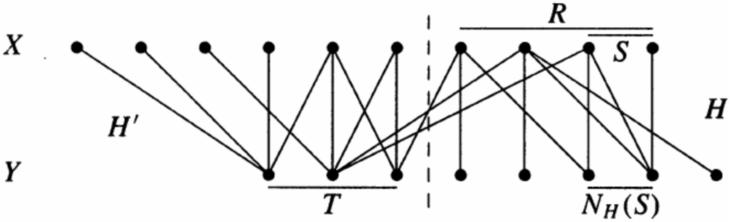


Given a minimum vertex cover Q,
we construct a matching M of the same size.



$$R = Q \cap X \qquad T = Q \cap Y$$

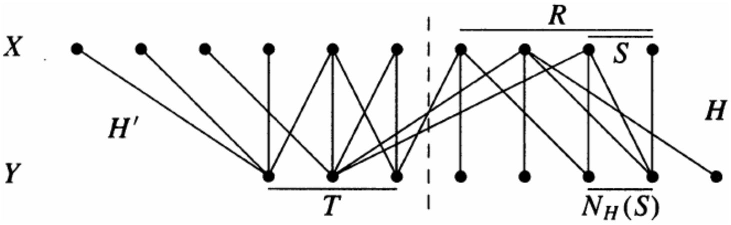
Given a minimum vertex cover Q,
we construct a matching M of the same size.



$$R = Q \cap X \qquad T = Q \cap Y$$

$$H \triangleq (R \cup (Y - T))\text{-induced subgraph of } G$$
$$H' \triangleq (T \cup (X - R))\text{-induced subgraph of } G$$

Given a minimum vertex cover Q,
we construct a matching M of the same size.

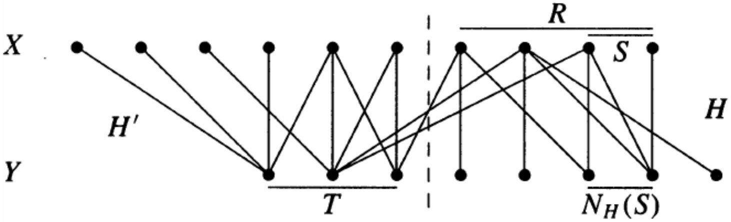


$$R = Q \cap X \qquad T = Q \cap Y$$

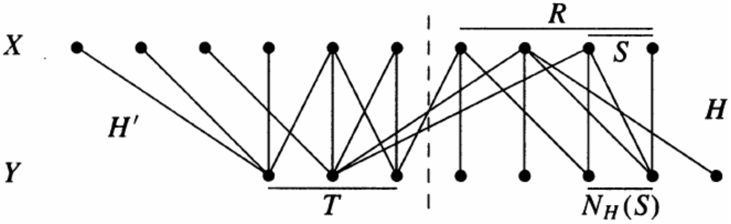
$$H \triangleq (R \cup (Y - T))\text{-induced subgraph of } G$$
$$H' \triangleq (T \cup (X - R))\text{-induced subgraph of } G$$

H has a R-perfect matching and H' has a T-perfect matching.

H has a R-perfect matching and H' has a T-perfect matching.

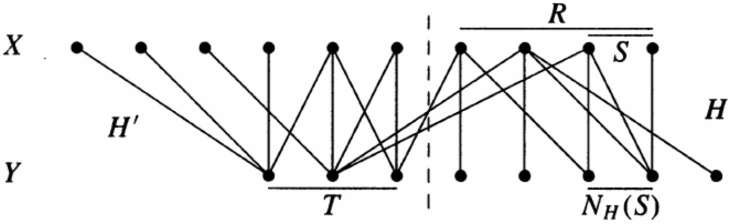


H has a R-perfect matching and H' has a T-perfect matching.



To show that Hall's Condition holds for H (and R).

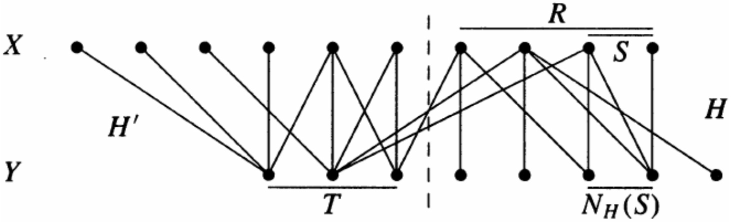
H has a R-perfect matching and H' has a T-perfect matching.



To show that Hall's Condition holds for H (and R).

By contradiction.

H has a R-perfect matching and H' has a T-perfect matching.

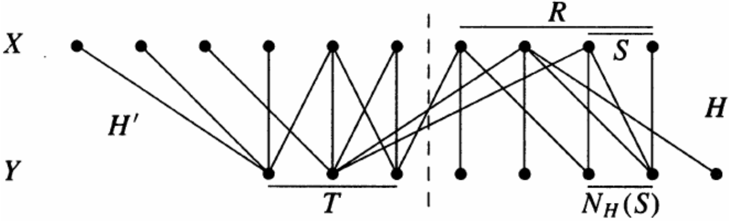


To show that Hall's Condition holds for H (and R).

By contradiction.

$$\exists S \subseteq R.\ \left|N_H(S)\right| < \left|S\right|$$

H has a R-perfect matching and H' has a T-perfect matching.



To show that Hall's Condition holds for H (and R).

By contradiction.

$$\exists S \subseteq R.\ \left|N_H(S)\right| < \left|S\right|$$

$T \cup (R - S + N_H(S))$ is a smaller vertex cover than Q.

Augmenting Path Algorithm for Maximum Bipartite Matching

Augmenting Path Algorithm for Maximum Bipartite Matching

- $M = \emptyset$

Augmenting Path Algorithm for Maximum Bipartite Matching

- $M = \emptyset$
- **While** there is an M-augmenting path P:
  - Enlarge M by P: $M \leftarrow M \oplus P$

Augmenting Path Algorithm for Maximum Bipartite Matching

- $M = \emptyset$
- **While** there is an M-augmenting path P:
  - Enlarge M by P: $M \leftarrow M \oplus P$
- **Return** M

Augmenting Path Algorithm for Maximum Bipartite Matching

- $M = \emptyset$
- **While** there is an M-augmenting path P:
  - Enlarge M by P: $M \leftarrow M \oplus P$
- **Return** M

How to find M-augmenting paths efficiently?

Augmenting Path Algorithm for Maximum Bipartite Matching

- $M = \emptyset$
- **While** there is an M-augmenting path P:
  - Enlarge M by P: $M \leftarrow M \oplus P$
- **Return** M

How to find M-augmenting paths efficiently?

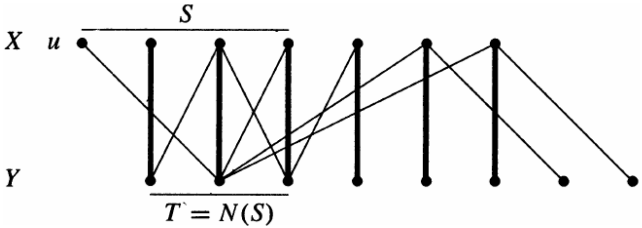
How to prove that the M returned is a maximum matching?

Augmenting Path Algorithm for Maximum Bipartite Matching

- $M = \emptyset$
- **While** there is an M-augmenting path P:
  - Enlarge M by P: $M \leftarrow M \oplus P$
- **Return** M

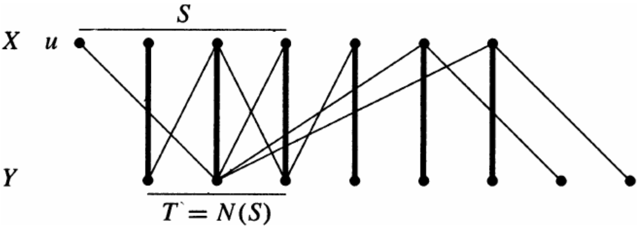
Augmenting Path Algorithm for Maximum Bipartite Matching

- $M = \emptyset$
- **While** there is an M-augmenting path P:
  - Enlarge M by P: $M \leftarrow M \oplus P$
- **Return** M



Augmenting Path Algorithm for Maximum Bipartite Matching

- $M = \emptyset$
- **While** there is an M-augmenting path P:
  - Enlarge M by P: $M \leftarrow M \oplus P$
- **Return** M



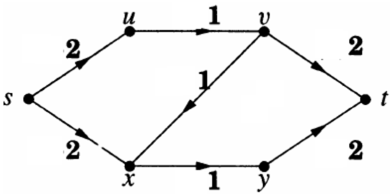
When the algorithm terminates,

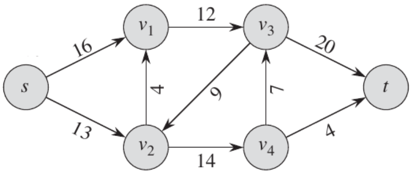
it actually finds a vertex cover $T \cup (X - S)$ of size M.

Definition (Network (网络))

A network is a digraph with

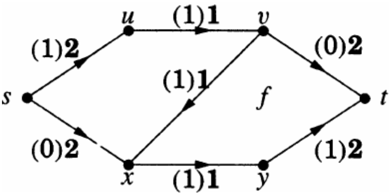
- a distinguished source vertex s,
- a distinguished sink vertex t,
- a capacity $c(e) \geq 0$ on each edge e

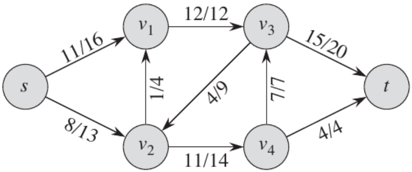




Definition (Flow (流))

A flow f is a function that assigns a value $f(e)$ to each edge e.





Definition (Feasible Flow (可行流))

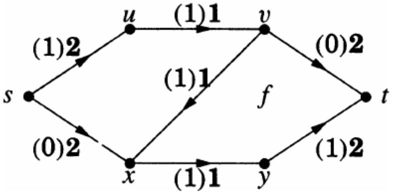
A flow f is feasible if it satisfies

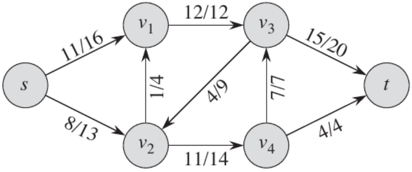
Capacity Constraints:

$$\forall e \in E\ 0 \le f(e) \le c(e)$$

Flow Conservation:

$$\forall v \in V \setminus \{s,t\}\ f^+(v) = f^-(v)$$



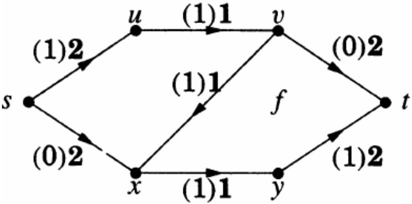


$$f^+(v) = \sum_{(v,w)\in E} f(v,w) \qquad f^-(v) = \sum_{(u,v)\in E} f(u,v)$$

Definition (Value (值))

The value $\mathrm{val}(f)$ of a flow f is

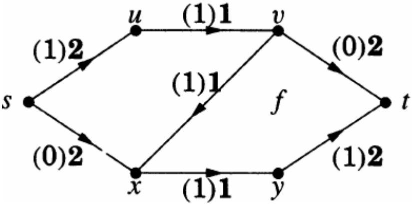
$$\mathrm{val}(f) = f^{-}(t) = f^{+}(s).$$

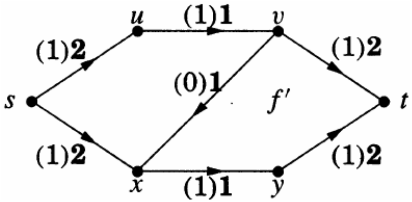


Definition (Value (值))

The value $\mathsf{val}(f)$ of a flow f is

$$\mathsf{val}(f) = f^-(t) = f^+(s).$$

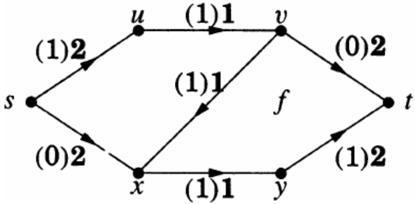


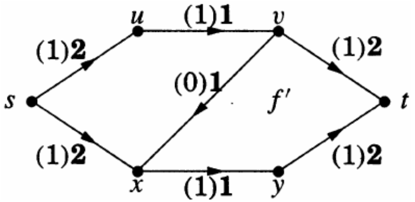


Definition (Value (值))

The value $\mathsf{val}(f)$ of a flow f is

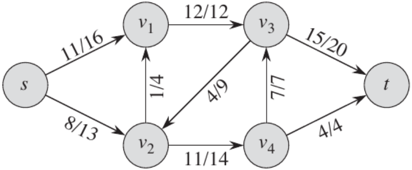
$$\mathsf{val}(f) = f^-(t) = f^+(s).$$



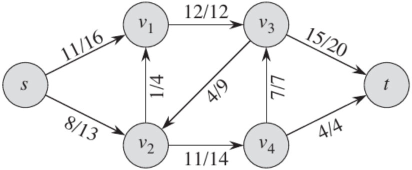


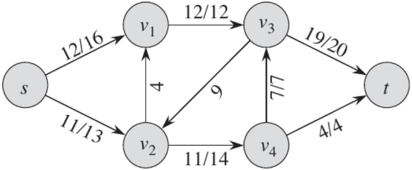
Definition (Maximum Flow (最大流))

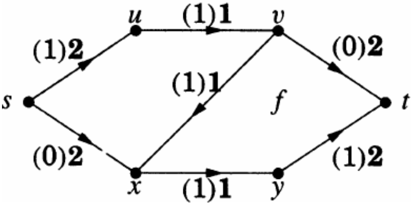
A maximum flow is a feasible flow of maximum value.



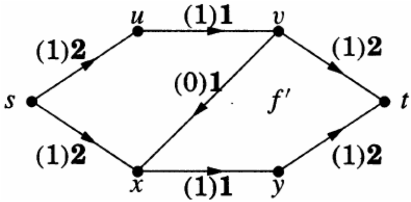
s
v_1
v_2
v_3
v_4
t
11/16
8/13
12/12
1/4
4/9
7/7
11/14
15/20
4/4


s
v_1
v_2
v_3
v_4
t
11/16
8/13
12/12
1/4
4/9
7/7
15/20
11/14
4/4

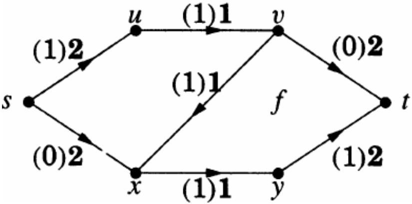

s
v_1
v_2
v_3
v_4
t
12/16
11/13
12/12
4
9
7/7
19/20
11/14
4/4

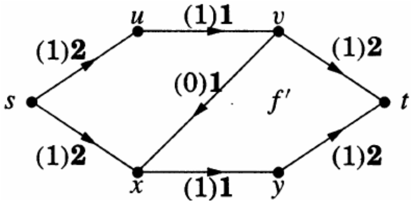


u
(1)1
v
(1)2
(0)2
(1)1
s
f
t
(0)2
(1)2
x
(1)1
y

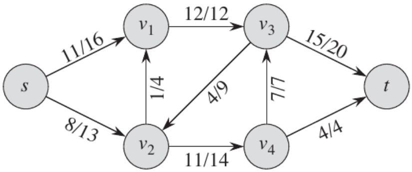


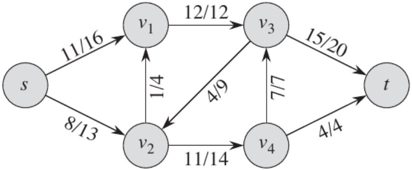
u
(1)1
v
(1)2
(1)2
(0)1
s
f′
t
(1)2
(1)2
x
(1)1
y

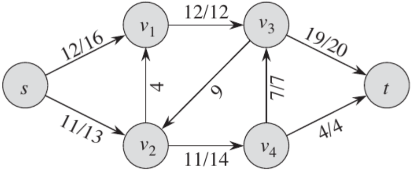


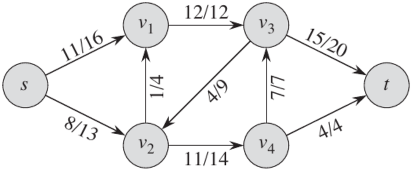


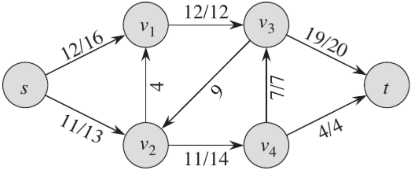
$$s \sim t \text{ path} : s - x - v - t$$


s
t
v1
v2
v3
v4
11/16
8/13
12/12
1/4
4/9
7/7
15/20
4/4
11/14

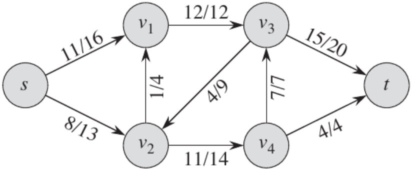

s
t
v_1
v_2
v_3
v_4
11/16
8/13
12/12
1/4
4/9
7/7
11/14
15/20
4/4

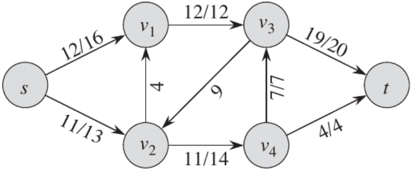

s
t
v_1
v_2
v_3
v_4
12/16
11/13
12/12
4
9
7/7
11/14
19/20
4/4





$$s \sim t \text{ path} : s - v_1 - v_2 - v_3 - t$$

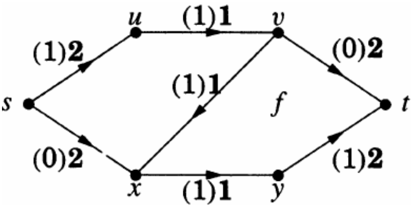


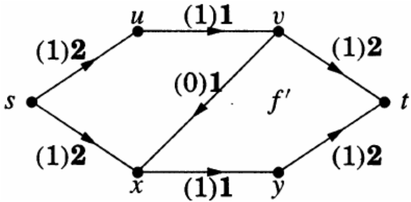


$$s \sim t \text{ path} : s - v_1 - v_2 - v_3 - t$$

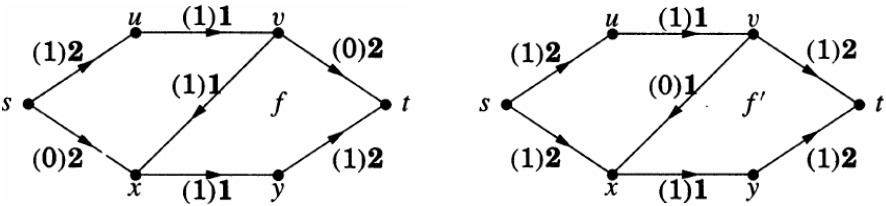
$$s \sim t \text{ path} : s - v_2 - v_3 - t$$

$$s \sim t \text{ path} : s - x - v - t$$





$$s \sim t \text{ path} : s - x - v - t$$



Definition (f-augmenting Paths (增广路径))

When f is a feasible flow, an f-augmenting path is a $s \sim t$ path P in the underlying graph such that for each edge $e \in E(P)$,

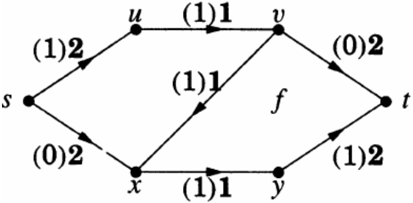
(a) if P follows e in the forward direction, then $f(e) < c(e)$;

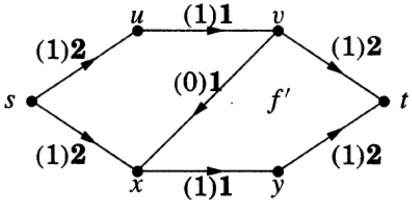
(b) if P follows e in the backward direction, then $f(e) > 0$.

Definition (f-augmenting Paths)

Let P be an f-augmenting path.

$$\epsilon(e) = \begin{cases} c(e) - f(e) & \text{if } e \text{ is forward on } P \\ f(e) & \text{if } e \text{ is backward on } P \end{cases}$$

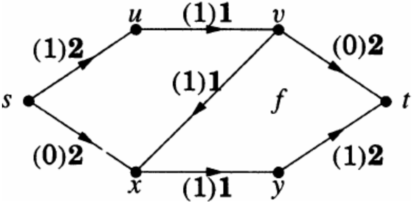


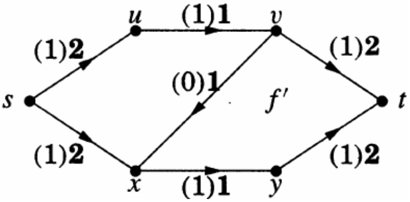


Definition (f-augmenting Paths)

Let P be an f-augmenting path.

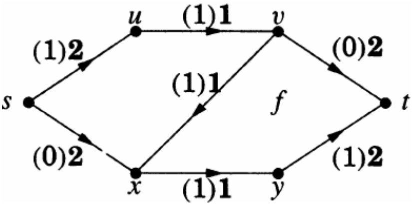
$$\epsilon(e) = \begin{cases} c(e) - f(e) & \text{if } e \text{ is forward on } P \\ f(e) & \text{if } e \text{ is backward on } P \end{cases}$$



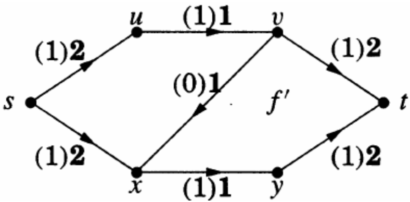


An f-augmenting path leads to a flow with larger value.

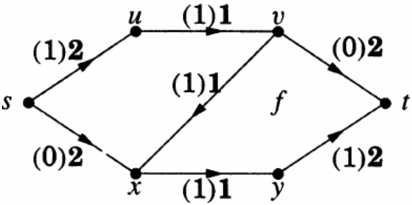
tolerance of P : $\min\limits_{e \in E(P)} \epsilon(e)$



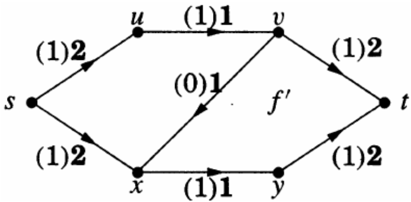
u
(1)1
v
(1)2
(0)2
(1)1
s
f
t
(0)2
(1)2
x
(1)1
y



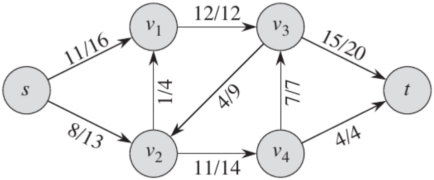
u
(1)1
v
(1)2
(1)2
(0)1
s
f'
t
(1)2
(1)2
x
(1)1
y



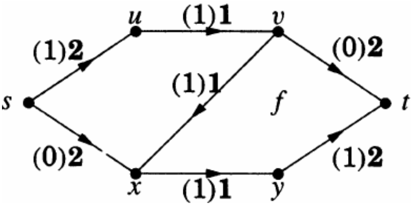
u
(1)1
v
(1)2
(0)2
(1)1
s
f
t
(0)2
(1)2
x
(1)1
y

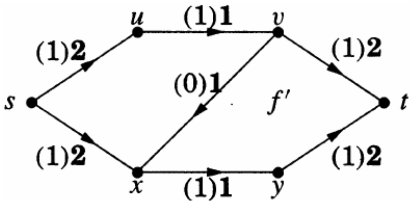


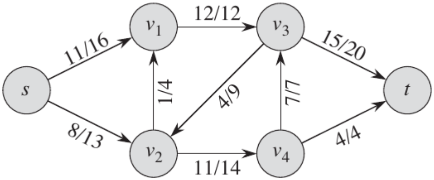
u
(1)1
v
(1)2
(1)2
(0)1
s
f′
t
(1)2
(1)2
x
(1)1
y

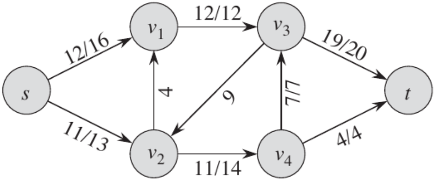


12/12
11/16
15/20
1/4
4/9
7/7
8/13
4/4
11/14


u
(1)1
v
(1)2
(0)2
(1)1
s
f
t
(0)2
(1)2
x
(1)1
y


u
(1)1
v
(1)2
(1)2
(0)1
s
f′
t
(1)2
(1)2
x
(1)1
y

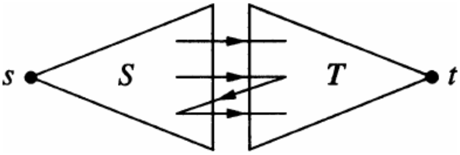

v1
12/12
v3
11/16
15/20
s
1/4
4/9
7/7
t
8/13
v2
v4
4/4
11/14

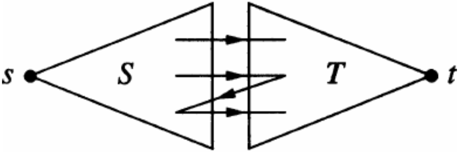

v1
12/12
v3
12/16
19/20
s
4
9
7/7
t
11/13
v2
v4
4/4
11/14

Definition (Source/Sink Cut (割))

In a network, a source/sink cut $[S, T]$ consists of the edges from a source set S to a sink set T, where

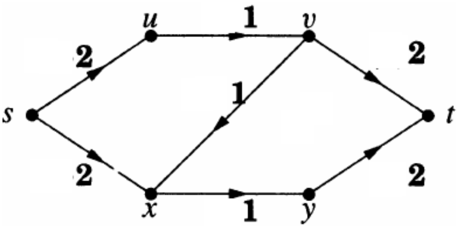
$$(T = V \setminus S) \wedge (s \in S) \wedge (t \in T)$$

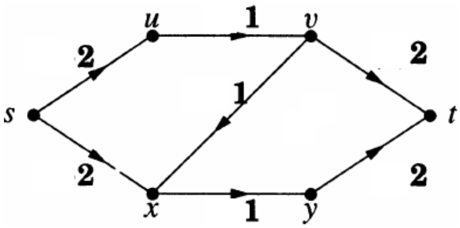




Definition (Capacity of Cut (割的容量))

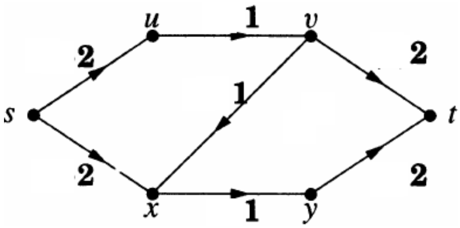
$$\mathsf{cap}(S,T) = \sum_{u \in S, v \in T, uv \in E} c(u,v)$$


u
1
v
2
2
1
s
t
2
2
x
1
y



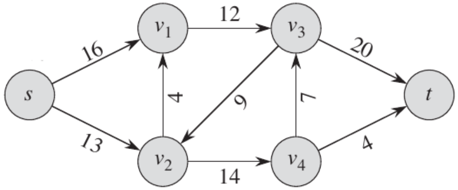
Definition (Minimum Cut (最小割))

A minimum cut is a cut of minimum value.



Definition (Minimum Cut (最小割))

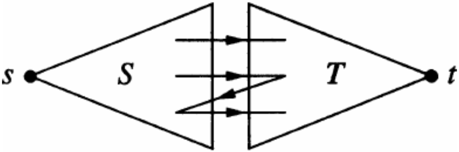
A minimum cut is a cut of minimum value.

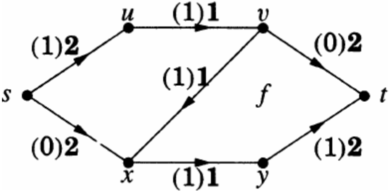


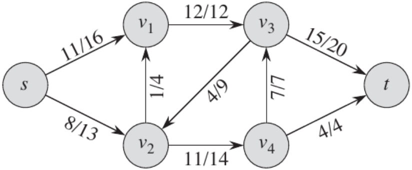
Theorem (Weak Duality (弱对偶定理))

Let f be any feasible flow and $[S, T]$ be any source/sink cut.

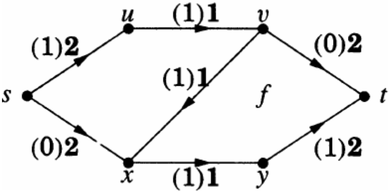
$$\mathit{val}(f) \leq \mathit{cap}(S, T).$$

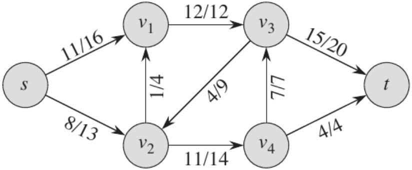




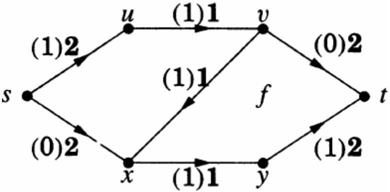


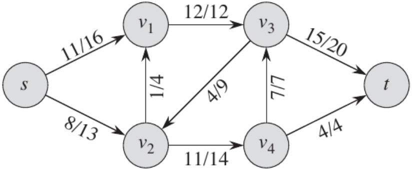
$$f^+(U) = \sum_{u \in U, v \in \overline{U}, (u,v) \in E} f(u,v)$$





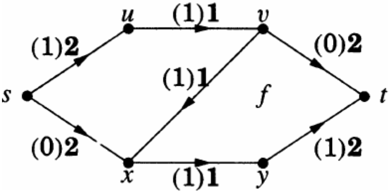
$$f^+(U) = \sum_{u \in U, v \in \overline{U}, (u,v) \in E} f(u, v) \qquad f^-(U) = \sum_{v \in \overline{U}, u \in U, (v,u) \in E} f(v, u)$$

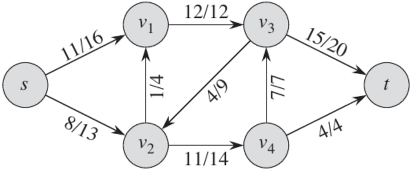




$$f^{+}(U) = \sum_{u \in U, v \in \overline{U}, (u,v) \in E} f(u,v) \qquad f^{-}(U) = \sum_{v \in \overline{U}, u \in U, (v,u) \in E} f(v,u)$$

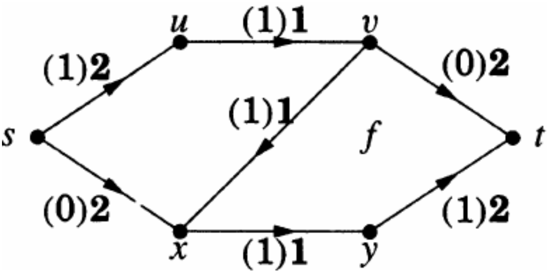
net flow out of U: $f^{+}(U) - f^{-}(U) =$

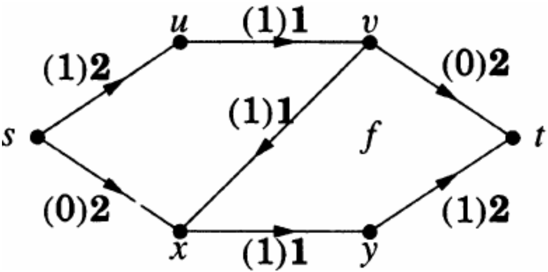




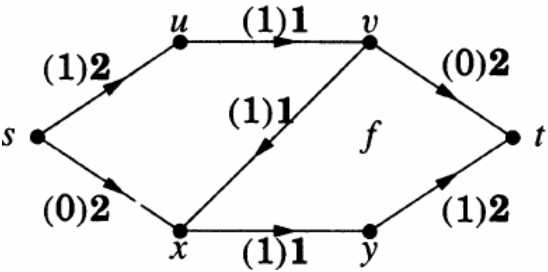
$$f^+(U) = \sum_{u\in U, v\in \overline{U}, (u,v)\in E} f(u,v) \qquad f^-(U) = \sum_{v\in \overline{U}, u\in U, (v,u)\in E} f(v,u)$$

net flow out of U: $f^+(U) - f^-(U) = \sum_{u\in U}(f^+(u) - f^-(u))$


u
(1)1
v
(1)2
(0)2
(1)1
s
f
t
(0)2
(1)2
x
(1)1
y

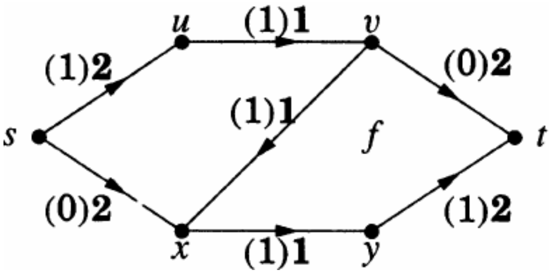


$$\forall U \subseteq (V - \{s, t\})\ f^+(U) = f^-(U)$$



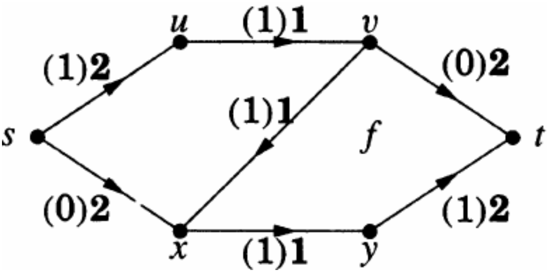
$$\forall U \subseteq (V - \{s,t\})\ f^{+}(U) = f^{-}(U)$$

$$s \in U \wedge t \notin U \implies f^{+}(U) - f^{-}(U) =$$



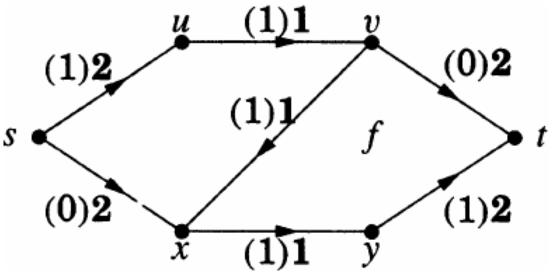
$$\forall U \subseteq (V - \{s, t\})\ f^{+}(U) = f^{-}(U)$$

$$s \in U \wedge t \notin U \implies f^{+}(U) - f^{-}(U) = f^{+}(s) =$$



$$\forall U \subseteq (V - \{s,t\})\ f^+(U) = f^-(U)$$

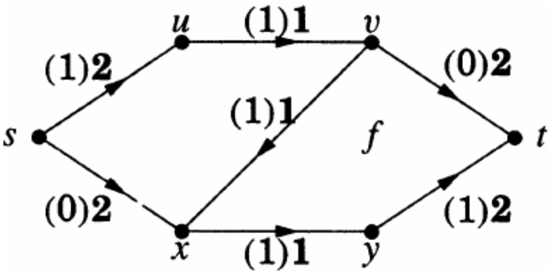
$$s \in U \wedge t \notin U \implies f^+(U) - f^-(U) = f^+(s) = \mathsf{val}(f)$$



$$\forall U \subseteq (V - \{s, t\})\ f^+(U) = f^-(U)$$

$$s \in U \wedge t \notin U \implies f^+(U) - f^-(U) = f^+(s) = \mathsf{val}(f)$$

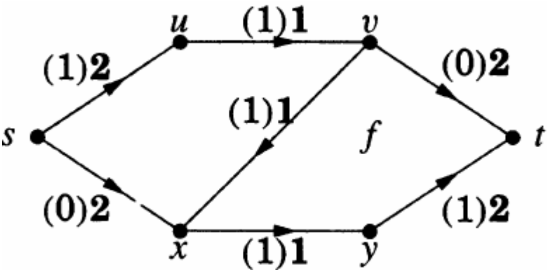
$$s \notin U \wedge t \in U \implies f^+(U) - f^-(U) =$$



$$\forall U \subseteq (V - \{s,t\})\ f^+(U) = f^-(U)$$

$$s \in U \wedge t \notin U \implies f^+(U) - f^-(U) = f^+(s) = \mathsf{val}(f)$$

$$s \notin U \wedge t \in U \implies f^+(U) - f^-(U) = f^-(t) =$$



$$\forall U \subseteq (V - \{s,t\})\ f^+(U) = f^-(U)$$

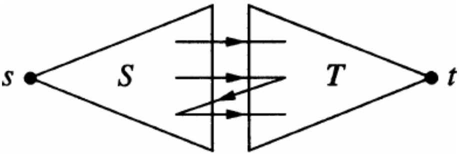
$$s \in U \wedge t \notin U \implies f^+(U) - f^-(U) = f^+(s) = \mathsf{val}(f)$$

$$s \notin U \wedge t \in U \implies f^+(U) - f^-(U) = f^-(t) = \mathsf{val}(f)$$

Theorem (Weak Duality (弱对偶定理))

Let f be any feasible flow and $[S,T]$ be any source/sink cut.

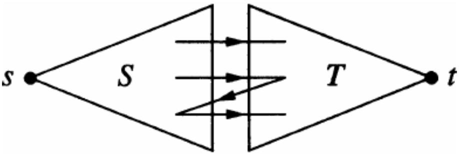
$$\mathit{val}(f) \leq \mathit{cap}(S,T).$$



Theorem (Weak Duality (弱对偶定理))

Let f be any feasible flow and $[S, T]$ be any source/sink cut.

$$\mathit{val}(f) \leq \mathit{cap}(S, T).$$

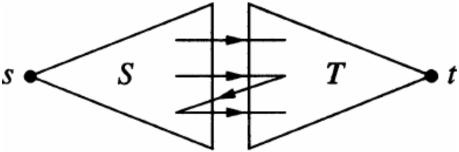


$$\text{val}(f) = f^{+}(S) - f^{-}(S)$$

Theorem (Weak Duality (弱对偶定理))

Let f be any feasible flow and $[S,T]$ be any source/sink cut.

$$\mathit{val}(f) \le \mathit{cap}(S,T).$$

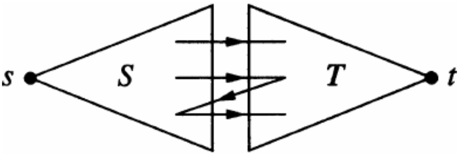


$$\mathrm{val}(f) = f^+(S) - f^-(S) \le f^+(S)$$

Theorem (Weak Duality (弱对偶定理))

Let f be any feasible flow and $[S,T]$ be any source/sink cut.

$$\mathit{val}(f) \le \mathit{cap}(S,T).$$



$$\text{val}(f) = f^{+}(S) - f^{-}(S) \le f^{+}(S) \le \text{cap}(S,T)$$

Lemma

$$\max_f \mathit{val}(f) \le \min_{[S,T]} \mathit{cap}(S,T)$$

Lemma

$$\max_{f} \mathit{val}(f) \leq \min_{[S,T]} \mathit{cap}(S,T)$$

What if $\mathsf{val}(f) = \mathsf{cap}(S,T)$ for some flow f and some cut $[S,T]$?

Lemma

$$\max_{f} \mathit{val}(f) \leq \min_{[S,T]} \mathit{cap}(S,T)$$

What if $\mathsf{val}(f) = \mathsf{cap}(S,T)$ for some flow f and some cut $[S,T]$?

f is maximum and $[S,T]$ is minimum

$$\mathsf{val}(f) = f^{+}(S) - f^{-}(S) = f^{+}(S) = \mathsf{cap}(S, T)$$

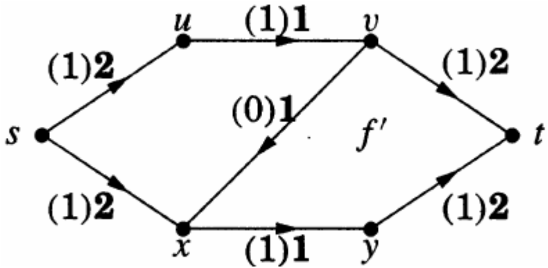
$$\mathsf{val}(f) = f^{+}(S) - f^{-}(S) = f^{+}(S) = \mathsf{cap}(S,T)$$

$$\boxed{f^{-1}(S) = 0 \wedge f^{+}(S) = \mathsf{cap}(S,T)}$$

$$\mathsf{val}(f) = f^{+}(S) - f^{-}(S) = f^{+}(S) = \mathsf{cap}(S,T)$$

$$\boxed{f^{-1}(S) = 0 \land f^{+}(S) = \mathsf{cap}(S,T)}$$

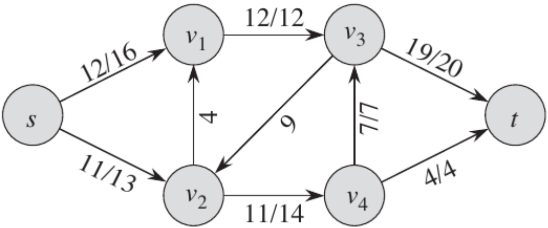
$$\boxed{\forall e \in [T,S]\ f(e) = 0 \land \forall e \in [S,T]\ f(e) = \mathsf{cap}(e)}$$



$$\mathsf{val}(f) = f^+(S) - f^-(S) \textcolor{red}{=} f^+(S) \textcolor{red}{=} \mathsf{cap}(S,T)$$

$$\boxed{f^{-1}(S) = 0 \wedge f^+(S) = \mathsf{cap}(S,T)}$$

$$\boxed{\forall e \in [T,S]\ f(e) = 0 \wedge \forall e \in [S,T]\ f(e) = \mathsf{cap}(e)}$$



Theorem (Max-flow Min-cut Theorem (Ford and Fulkerson; 1956))

$$\max_{f} val(f) = \min_{[S,T]} cap(S,T)$$

(Strong Duality)

L. R. Ford Jr. $(1927 \sim 2017)$

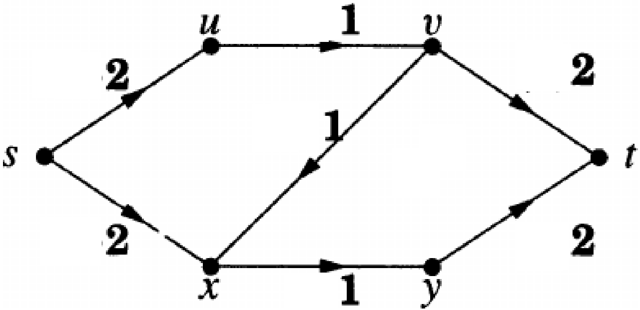
D. R. Fulkerson $(1924 \sim 1976)$

The Ford-Fulkerson Method

Repeatedly finding f-augmenting paths until no more ones exist.

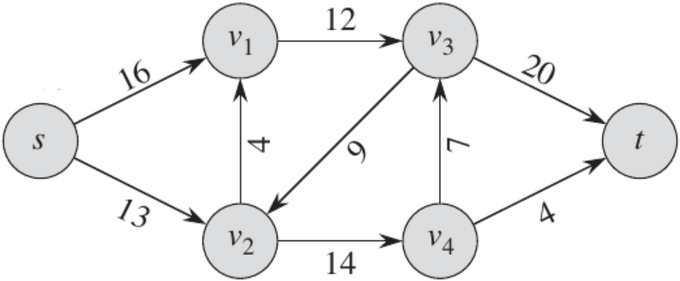
The Ford-Fulkerson Method

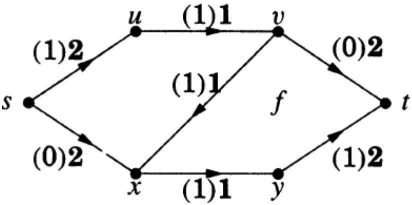
Repeatedly finding f-augmenting paths until no more ones exist.

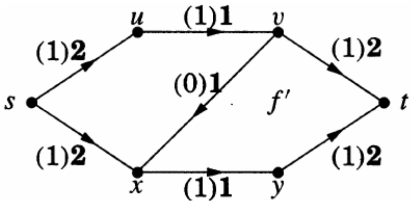


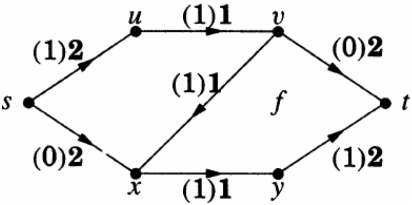
The Ford-Fulkerson Method

Repeatedly finding f-augmenting paths until no more ones exist.

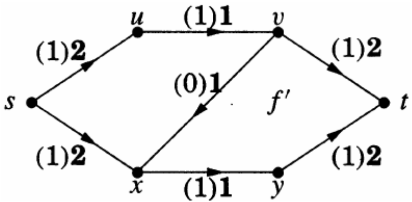



u
(1)1
v
(1)2
(0)2
(1)1
s
f
t
(0)2
(1)2
x
(1)1
y

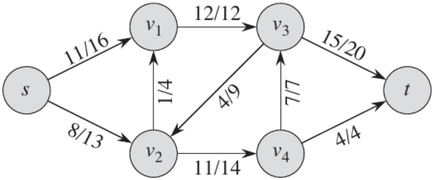

u
(1)1
v
(1)2
(1)2
(0)1
s
f'
t
(1)2
(1)2
x
(1)1
y



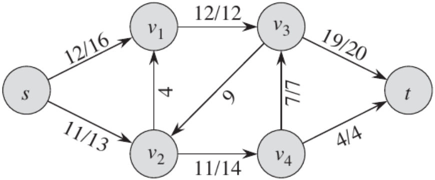
u
(1)1
v
(1)2
(0)2
(1)1
f
s
t
(0)2
(1)2
x
(1)1
y



u
(1)1
v
(1)2
(1)2
(0)1
f′
s
t
(1)2
(1)2
x
(1)1
y



v1
12/12
v3
11/16
15/20
s
1/4
4/9
7/7
t
8/13
4/4
v2
11/14
v4



v1
12/12
v3
12/16
19/20
s
4
9
7/7
t
11/13
4/4
v2
11/14
v4

Theorem

A feasible flow f is maximum iff there are no f-augmenting paths.

Theorem

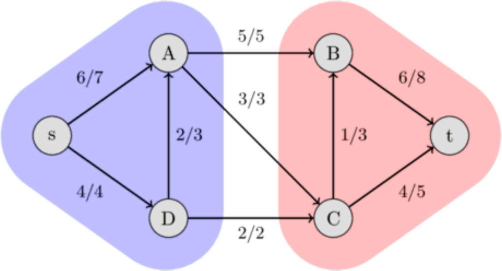
A feasible flow f is maximum iff there are no f-augmenting paths.

We construct a cut $[S, T]$ with $\mathsf{val}(f) = \mathsf{cap}(S, T)$.

Theorem

A feasible flow f is maximum iff there are no f-augmenting paths.

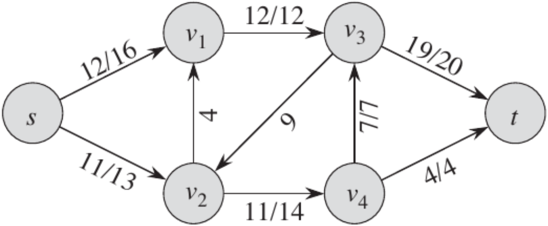
We construct a cut $[S, T]$ with $\mathsf{val}(f) = \mathsf{cap}(S, T)$.



$S \triangleq \{$the vertices reachable from s along partial f-augmenting paths$\}$

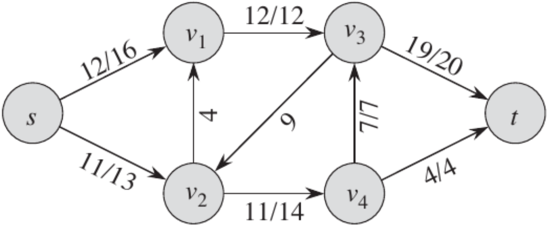
$$S \triangleq \{\text{the vertices reachable from } s \text{ along } \textcolor{red}{\text{partial}}\ f\text{-augmenting paths}\}$$

$$T \triangleq V - S$$



$$S \triangleq \{\text{the vertices reachable from } s \text{ along partial } f\text{-augmenting paths}\}$$

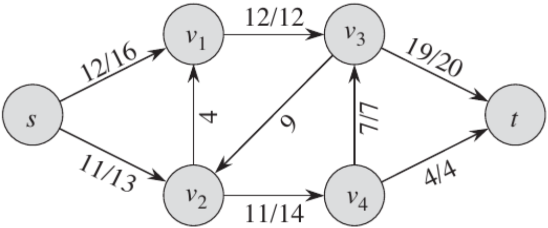
$$T \triangleq V - S$$



$$f^{-1}(S) = 0 \wedge f^{+}(S) = \mathsf{cap}(S, T)$$

$S \triangleq$ {the vertices reachable from s along partial f-augmenting paths}

$$T \triangleq V - S$$



$$f^{-1}(S) = 0 \wedge f^{+}(S) = \mathsf{cap}(S, T)$$

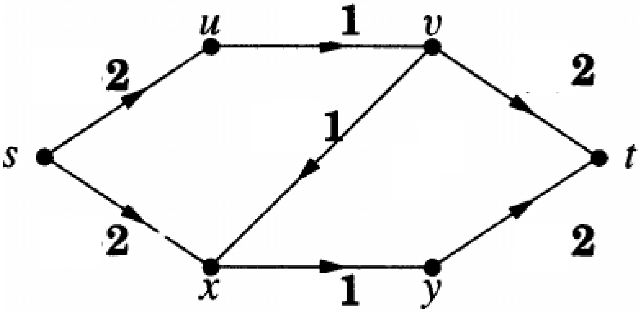
$$\mathsf{val}(f) = f^{+}(S) - f^{-}(S) = f^{+}(S) = \mathsf{cap}(S, T)$$

The Edmonds-Karp Algorithm

Using BFS (Breadth-first Search) to find f-augmenting paths.

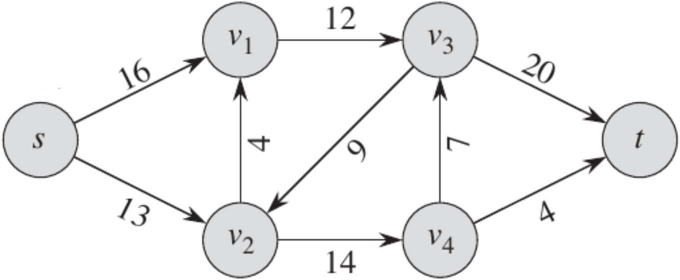
The Edmonds-Karp Algorithm

Using BFS (Breadth-first Search) to find f-augmenting paths.



The Edmonds-Karp Algorithm

Using BFS (Breadth-first Search) to find f-augmenting paths.



Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph.

There is an X-perfect matching of G iff

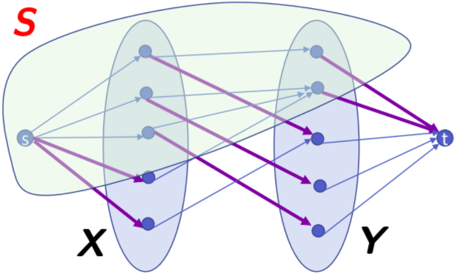
$$\forall W \subseteq X \; \Big|W\Big| \leq \Big|N_G(W)\Big|$$

Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph.

There is an X-perfect matching of G iff

$$\forall W \subseteq X \ \left|W\right| \leq \left|N_G(W)\right|$$



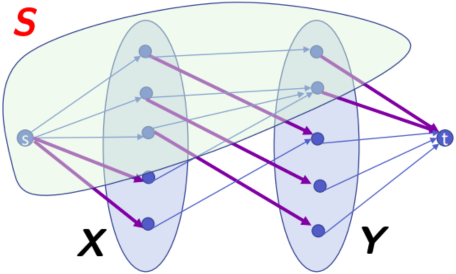
$\forall x \in X \ c(s,x) = 1 \quad \forall y \in Y \ c(y,t) = 1 \quad \forall x \in X, y \in Y \ c(x,y) = \infty$

Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph.

There is an X-perfect matching of G iff

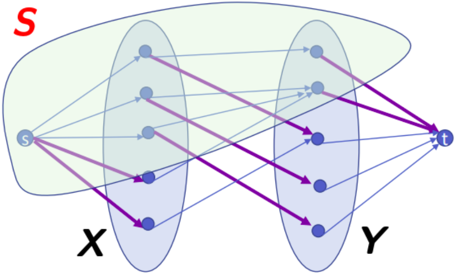
$$\forall W \subseteq X \ \left|W\right| \le \left|N_G(W)\right|$$



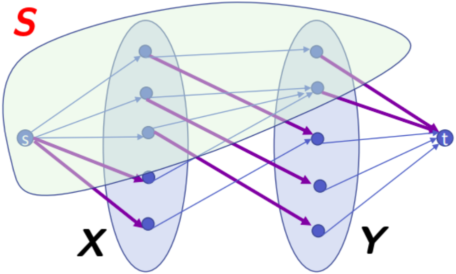
$$\forall x \in X \ c(s, x) = 1 \quad \forall y \in Y \ c(y, t) = 1 \quad \forall x \in X, y \in Y \ c(x, y) = \infty$$

We need to show that $\max\limits_f \mathsf{val}(f) = \left|X\right|$.

We need to show that $\min_{[S,\overline{S}]} \mathsf{cap}(S, \overline{S}) = \left| X \right|.$

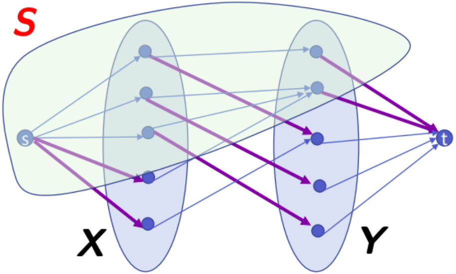


We need to show that $\min_{[S,\overline{S}]} \mathsf{cap}(S, \overline{S}) = \left|X\right|$.



$$\min_{[S,\overline{S}]} \mathsf{cap}(S, \overline{S}) \leq \left|X\right|$$

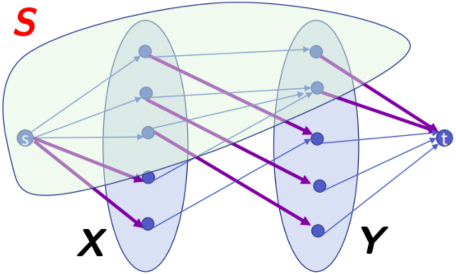
We need to show that $\min_{[S,\overline{S}]} \mathsf{cap}(S,\overline{S}) = \left|X\right|$.



$$\min_{[S,\overline{S}]} \mathsf{cap}(S,\overline{S}) \leq \left|X\right|$$

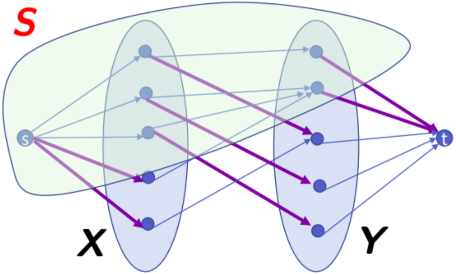
Therefore, we need to show that $\min_{[S,\overline{S}]} \mathsf{cap}(S,\overline{S}) \geq \left|X\right|$.

Let $[S, \overline{S}]$ be a cut. We need to show that $\mathsf{cap}(S, \overline{S}) \geq \left|X\right|$.



$$\boxed{N(S \cap X) \not\subseteq (S \cap Y)}$$

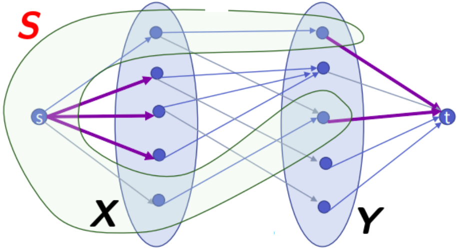
Let $[S, \overline{S}]$ be a cut. We need to show that $\mathsf{cap}(S, \overline{S}) \geq \left|X\right|$.



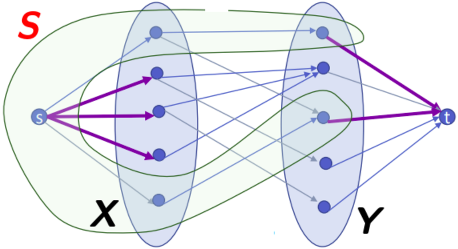
$$\boxed{N(S \cap X) \not\subseteq (S \cap Y)}$$

$$\mathsf{cap}(S, \overline{S}) = \infty \geq \left|X\right|$$

$$\boxed{N(S \cap X) \subseteq (S \cap Y)}$$

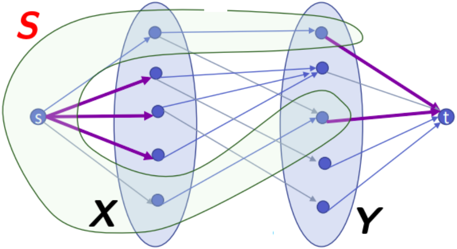


$$\boxed{N(S \cap X) \subseteq (S \cap Y)}$$



$$\mathsf{cap}(S, \overline{S}) = \sum_{u \in S, v \in \overline{S}} c(x, y)$$

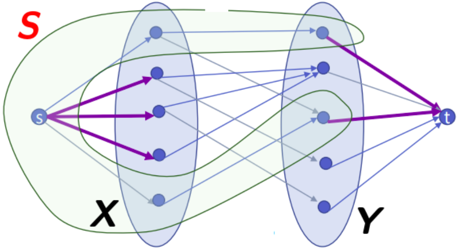
$$\boxed{N(S \cap X) \subseteq (S \cap Y)}$$



$$\mathrm{cap}(S, \overline{S}) = \sum_{u \in S, v \in \overline{S}} c(x, y)$$

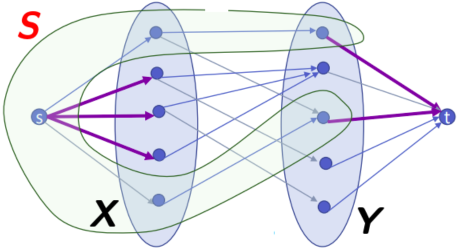
$$= \sum_{v \in \overline{S} \cap X} c(s, v) + \sum_{u \in S \cap Y} c(u, t)$$

$$\boxed{N(S \cap X) \subseteq (S \cap Y)}$$



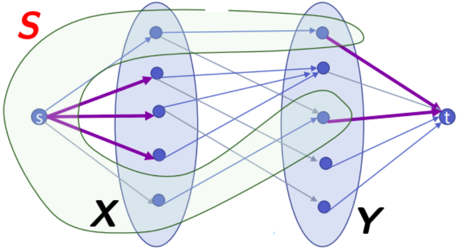
$$\begin{aligned}
\mathsf{cap}(S, \overline{S}) &= \sum_{u \in S, v \in \overline{S}} c(x, y) \\
&= \sum_{v \in \overline{S} \cap X} c(s, v) + \sum_{u \in S \cap Y} c(u, t) \\
&= \left|X\right| - \left|S \cap X\right| + \left|S \cap Y\right|
\end{aligned}$$

$$\boxed{N(S \cap X) \subseteq (S \cap Y)}$$



$$\begin{aligned}
\mathsf{cap}(S, \overline{S}) &= \sum_{u \in S, v \in \overline{S}} c(x, y) \\
&= \sum_{v \in \overline{S} \cap X} c(s, v) + \sum_{u \in S \cap Y} c(u, t) \\
&= \left|X\right| - \left|S \cap X\right| + \left|S \cap Y\right| \\
&\geq \left|X\right| - \left|S \cap X\right| + \left|N(S \cap X)\right|
\end{aligned}$$

$$\boxed{N(S \cap X) \subseteq (S \cap Y)}$$



$$\begin{aligned}
\text{cap}(S, \overline{S}) &= \sum_{u \in S, v \in \overline{S}} c(x, y) \\
&= \sum_{v \in \overline{S} \cap X} c(s, v) + \sum_{u \in S \cap Y} c(u, t) \\
&= \left|X\right| - \left|S \cap X\right| + \left|S \cap Y\right| \\
&\geq \left|X\right| - \left|S \cap X\right| + \left|N(S \cap X)\right| \geq \left|X\right|
\end{aligned}$$

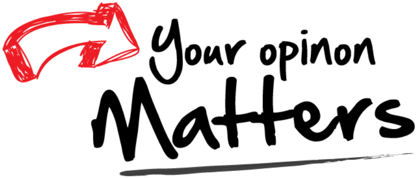
Theorem (König (1931), Egerváry (1931))

If G is a bipartite graph, then the maximum size of a mathching in G equals the minimum size of a vertex cover of G

Theorem (König (1931), Egerváry (1931))

If G is a bipartite graph, then the maximum size of a mathching in G equals the minimum size of a vertex cover of G


Thank You!



hfwei@hnu.edu.cn